Bijux Pollenomics: scientific evidence

Sources, method correspondence, predictions and model use

Bijux Pollenomics

30 September 2026

Source-bound research briefing

Evidence ladder

1. Archive/source custody and taxon-count mapping.
2. Author-equation and numerical implementation parity.
3. Observed-pollen held-out prediction.
4. LRA reconstruction and forward-pollen prediction.
5. Independent measured-vegetation validation: no admitted pair yet.

Each rung answers a different question; a result at one rung does not imply the next.

Evidence: `reports/README.md`; `reports/needed/README.md`

Four-country source and prediction accounting

Target	Sites	Samples	Assigned/native NISP	Scored/attempted bins
Denmark	1	91	122,998/129,768	24/24
Finland	28	1,797	1,229,019/1,260,281	435/435
Norway	52	3,544	2,018,295/2,286,020	1,084/1,178
Sweden	3	228	221,800/240,966	51/54

Closed 31-target author pollen composition. Excluded/unmapped mass remains outside the score. Zero Nordic samples are admitted to LRA by this study.

Evidence: `reports/science/denmark-finland-norway-author-pollen.md; reports/science/sweden-author-pollen.md`

Three further Swedish lake cores

- Metadata-only selection added Gilltjarnen, Lake Avegol and Flinkasjon outside the original Swedish targets.
- We assigned 231,115 of 238,797 native pollen grains to the 31 author targets; excluded and unmapped counts remain visible.
- All 63 site-bins received three fixed predictions, yielding 189 matched scores with no training-support refusal.
- Equal-site pollen MAE: IDW 0.017334, nearest site 0.017932, training mean 0.020682. Nearest site wins at Lake Avegol.

Three prior-exposed lake groups; direct observed pollen only, with no Nordic LRA or independent vegetation score.

Evidence: `reports/science/sweden-lake-transfer.md`; `docs/report/regions/nordic/swedish-lake-transfer/site-scores.csv`

Swedish training scope on shared pollen targets

- Hold out each Swedish site; train on the other two Swedish sites or those same two plus 81 DK/FI/NO sites.
- On 39 shared site-bins, inverse-distance-squared equal-site pollen MAE is 0.032846 versus 0.023226, favoring four-country training.
- Sweden-only refuses 15 of 54 candidate bins; four-country training predicts all 54.

Three previously exposed sites; most mean gain comes from Abborrtjarnen. No LRA or independent vegetation accuracy follows.

Evidence: `reports/science/sweden-training-scope.md`; `docs/report/regions/nordic/swedish-training-scope/paired-scores.csv`

Swedish source support changes the pollen score

- We froze 1,419 method predictions across 473 supported queries; 13 of 486 site-bin/policy queries refused.
- Adjacent nominal 500-year source bins and a nearest-five cap changed IDW error slightly and differently by site.
- Excluding source centers within 100 km raised matched equal-site IDW MAE from 0.023151 to 0.029171 on 52 bins.

Three exposed Swedish sites; site centers, no age posterior, LRA reconstruction or measured vegetation.

Evidence: `reports/science/sweden-spatiotemporal-sensitivity.md`; `docs/report/regions/nordic/swedish-support-sensitivity/paired-scores.csv`

What reproduces numerically

- Author raw-to-prepared replay: 17 example files, 871 raw sample rows, 15,028 matching prepared cells.
- WHH05: 301 raw grains map to 297 prepared; 31 paired means and SEs agree at 1500 BP.
- Reexecuted author R/native calculations on two grids (9+8 sites); 14,074 supported mean and SE values agree within declared tolerances.
- Archive TW2 differs: total variation 0.092/0.111; two taxa explain 73%/94% of each grid's L1 gap. Historical inputs remain unclosed.
- Archived ISTS and TBS mean arithmetic closes over all 10,500 grid-window rows, but biological classification and SE transformation remain open.

Evidence: `reports/science/publication-correspondence.md`; `reports/science/publication-correspondence/published-window-provenance.md`; `reports/science/publication-correspondence/author-example-reexecution.md`

LOVE optimizers: shared objectives, different searches

- We gave installed RcppDE and Python the same three declared pollen, deposition, background, PPE and bound profiles.
- Both attained objective gaps below 10^{-4} from independent optima; cross-evaluating selected candidates differed by less than 10^{-10}.
- Python reported success after 29–40 iterations and 600–1,230 evaluations. R used its full 400-iteration cap in all three cases.
- One objective has different, equally good vegetation vectors; vector equality would misstate solution quality.

Bounded objective and arithmetic checks; no identical random path, general global guarantee or ecological accuracy.

Evidence: `reports/science/method-correspondence.md`; `docs/report/method-certification/optimizer-solution-quality/summary.csv`

English LOVE: declared model profile

- 15 source sites, 1250–1750 BP, 30 targets; published PPE and fall speeds held fixed for retained taxa.
- Regional REVEALS plus ring-taxon-preserving LOVE; closed, nonnegative local cover.
- Uniform-start SLSQP: 300 iterations, 16,000 evaluations, objective tolerance 10^{-8}.
- 192/195 training fits complete; three failures also refused under a bounded 600-iteration check.

These are source-specific development choices, not universally optimal LRA parameters or the original R optimizer.

Evidence: `reports/science/english-lra-pollen-prediction.md`

English LOVE routes have different support

Route	Source axis and attempts	Retained result
Analytical	31 taxa; 15 folds	15/15 invalid compositions; all 54 local ring rows include negative cover.
Differential evolution	30 taxa; one pair, two requests	Both requests refused convergence; no held-out score.
Constrained SLSQP	30 taxa; 15 folds	14/15 folds predicted; BECKHAM had three training-fit refusals.

Excluding 159 Filipendula counts changed the taxon response. These development runs are not a paired solver ranking or a vegetation-accuracy comparison.

Evidence: `reports/science/english-lra-pollen-prediction.md`

Matched pollen comparison: result and limitation

	MAE		Hellinger	
	Direct	LRA	Direct	LRA
Training mean	.020496	.026273	.282082	.360671
Nearest	.019552	.021132	.255767	.286001
Inverse distance squared	.018579	.021088	.243673	.285415

14 paired English sites, 30 taxa; all three direct means remain lower after any one site is omitted. Prior exposure and unequal training information limit interpretation.

Evidence: `docs/report/pollen-reconstruction/author-lra-constrained/direct-pollen/paired-hellinger.json`

English basin radius: forward sensitivity only

- We replayed 42 frozen LRA vegetation estimates at 80%, 100% and 120% of each assumed target-lake radius.
- The actual forward consumer reproduced all 42 original pollen predictions exactly before target scoring.
- Inverse-distance-squared site-macro pollen MAE: .021032, .021088 and .021457 across those radii.
- On matched targets, direct pollen retains lower mean MAE and Hellinger for every method and radius factor.

14 prior-exposed English sites. Vegetation, regional cover and PPE were fixed; this does not calibrate basin geometry or admit Nordic LRA inputs.

Evidence: `docs/report/pollen-reconstruction/author-lra-constrained/radius-sensitivity/paired/report.json`; `reports/science/english-lra-pollen-prediction.md`

Model selection follows the target and support

- Fixed spatial baselines permit clear paired interpretation on observed pollen.
- A bounded RBF candidate improved one prior-exposed author comparison on common support but reduced coverage.
- Country-excluded tuning did not establish superiority over inverse-distance-squared in the Nordic development holdouts.
- A two-case synthetic PPE/transport winner reflects a transport-law mismatch and is not a real-core parameter recommendation.
- For LRA, input identity, chronology, basin geometry, PPE and denominator must be qualified before any parameter search is meaningful.

Evidence: `reports/science/model-use.md`; `reports/science/synthetic-model-selection/README.md`; `reports/needed/README.md`

HYDE and vegetation: retain the boundary

- Four-country HYDE source cropland grid sums match 84 country/year/scenario totals within 8.51×10^{-11} km^2.
- This checks modeled-source internal accounting, not pollen-model benefit.
- The retained release has conflicting rights and version declarations; formal CRS and historical calendar support remain unresolved.
- No independent vegetation measurement pair has been admitted for prediction accuracy.

Next experiments require source semantics and a matched ecological reference, with outcomes protected from model selection.

Evidence: `reports/science/hyde-cropland-source-account.md`; `reports/needed/README.md`

Why no vegetation-accuracy score exists yet

- EMPD2–NFI screen: 16 nearby candidates, zero measured pairs admitted for our benchmark.
- Färskesjön: the measured 7 m forest plot is 427 m from the lake pollen point; five volume classes do not supply 31-taxon landscape cover.
- Three Småland bogs: historical-map coverage and pollen anchors are published, but full numerical vegetation fractions are absent.
- A published southern-Swedish figure favors its authors' REVEALS estimates on nine rounded printed labels; it does not score our predictions or recover the full 26-taxon study.

A physical-site, time, taxon and landscape-support match is still required before ecological validation.

Evidence: `reports/science/vegetation-reference.md`; `docs/report/validation/empd2-nfi-pairing/report.json`

Open source decisions

- Abborrtjärnen: LandClim-selected age model, historical basin radius, eligible denominator, PPE use and rights.
- Bjäresjösjön: modern lake identity and nearby SEAD display intervals close; pollen-core identity and ecological event join remain refused.
- Solsø: 79 LandClim II samples versus 92 Neotoma source samples; the published age span retains 90, and no historical basin series is bound to the selected rows.
- Dalmutladdo (Norway): 72 source samples and 11 radiocarbon controls align; published 7.2 ha versus LandClim 8 ha, with no dated basin series. Zero LRA admission.
- Laihalampi (Finland): one LAIH core and 150 depths align; modern area claims agree conditionally, but selected ages and dated basin remain unqualified.
- Historical publication: exact source revisions and authenticated PFT/uncertainty transformation.
- Independent confirmation: new qualified cohort frozen before target inspection.

Evidence: `reports/needed/README.md`; `reports/science/solsoe-lra-input.md`; `reports/science/dalmutladdo-lra-input/README.md`; `reports/science/laihalampi-lra-input/README.md`